

CampusMind - AI Learning Assistant Solution for SDG 4: Quality Education

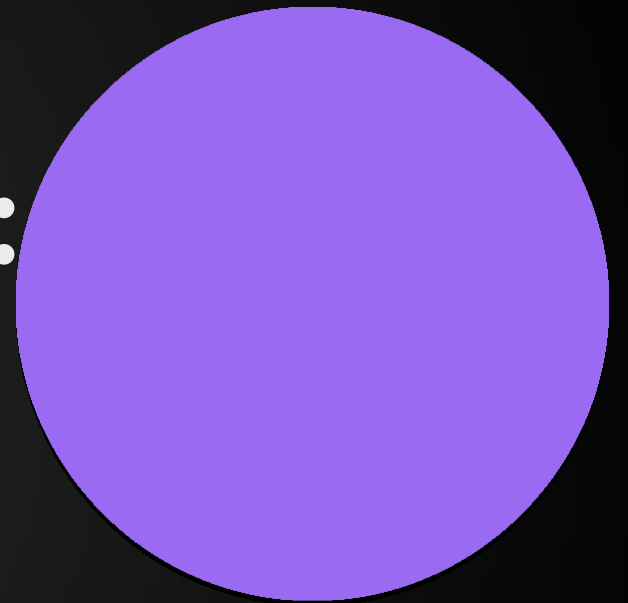
SUBJECT: LENOVO LEAP AI CAPSTONE PROJECT (MILESTONE 8)

SUBMITTED BY: PRATIK ASHWINI PANDEY

DATE: 08 JUNE 2026

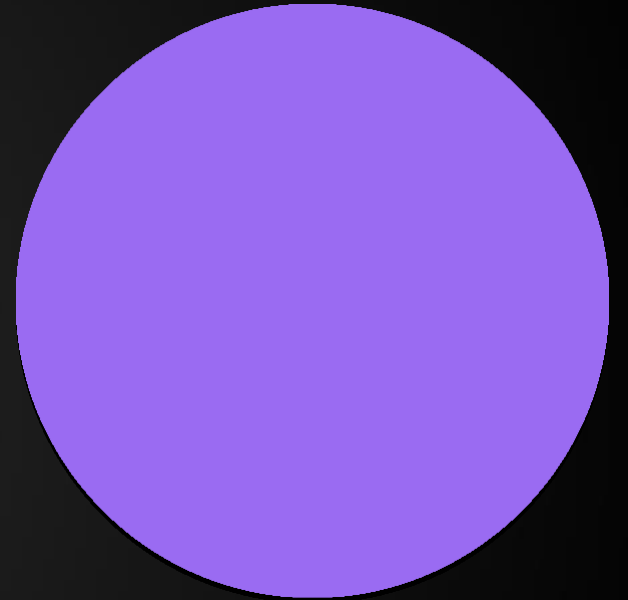
TOPIC: AI BASED CHATBOT

COURSE/BATCH: AI INTERNSHIP/JUNE 2026



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1. SDG & Problem Statement

 **Selected SDG:** Goal 4 - Quality Education

 **Why this SDG ?:** Many students, especially in rural areas, don't have access to personal tutors or quick doubt-solving support. They get stuck on concepts and have no one to ask.

 **Problem Statement:**

1. **What:** Students struggle to understand concepts because they have no one to ask when stuck
2. **Where:** Rural and semi-urban areas with limited access to coaching
3. **Who affected:** School and college students
4. **Why serious:** Without guidance, students lose interest or fall behind
5. **If not solved:** Education gap widens; more students rely on rote learning

2. Solution Description

 **Project Name:** CampusMind - AI Learning Assistant

 **How it works:**

1. Student types a question in the chat
2. AI understands the question and explains the concept in simple language
3. Student can ask follow-up questions
4. Chat history is maintained for context

 **Who will use it ?:** Students from Class 8 to college level

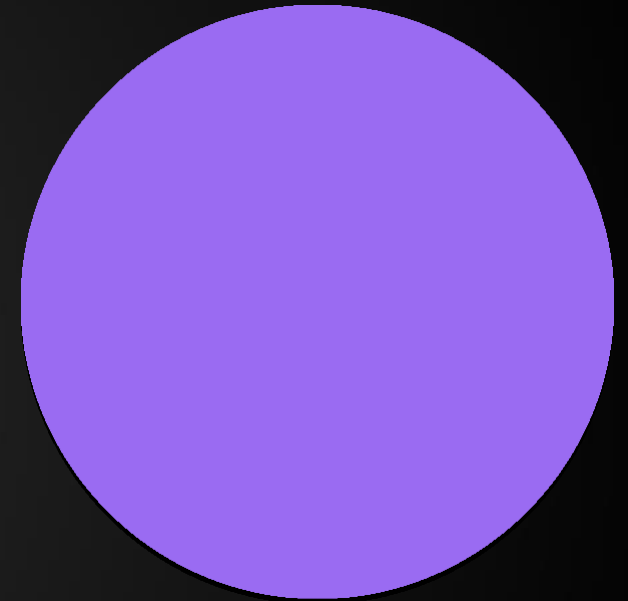
 **How it helps ?:** Provides instant, free, personalized learning support 24/7

3. Technology Stack

Component	Technology used
Frontend	Streamlit(via Python)
AI Model/API	Groq API
Programming Language	Python
Deployment	Hugging Face Spaces
Version Control	GitHub

4. Results (Code Snippet 1)

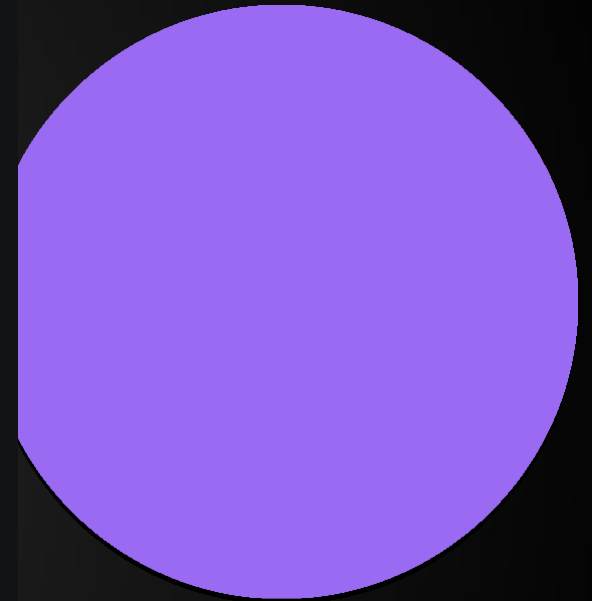
```
E: > edubuddy-ai-chatbot-main > app.py
1  # CampusMind - AI Learning Assistant (Powered by Groq)
2  # SDG 4: Quality Education
3
4  import streamlit as st
5  from groq import Groq
6  import os
7
8  # Page configuration
9  st.set_page_config(
10     page_title="CampusMind",
11     page_icon="🎓",
12     layout="wide"
13 )
14
15 # Custom Styling
16 st.markdown("""
17 <style>
18 .main {
19     padding-top: 1rem;
20 }
21
22 .main-header {
23     text-align: center;
24     padding: 20px;
25     border-radius: 15px;
26     background: linear-gradient(90deg, #4F46E5, #06B6D4);
27     color: white;
28     margin-bottom: 20px;
29 }
30
```



4. Results (Code Snippet 2)

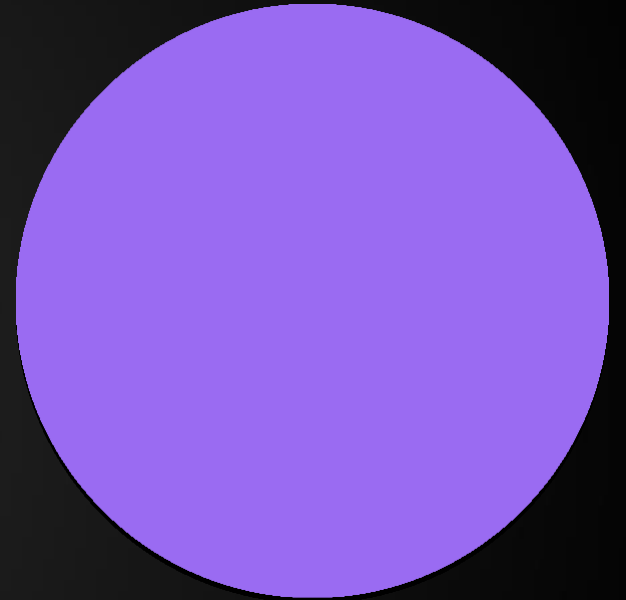
E: > edubuddy-ai-chatbot-main > app.py > ...

```
69 # Initialize chat history
70 if "messages" not in st.session_state:
71     st.session_state.messages = []
72
73 # Sidebar
74 with st.sidebar:
75     st.title("📖 About")
76
77     st.write("""
78     **CampusMind** is an AI-powered educational assistant
79     designed to help students understand concepts,
80     solve doubts, and learn more effectively.
81     """)
82
83     st.divider()
84
85     st.metric("💬 Messages", len(st.session_state.messages))
86
87     st.divider()
88
89     st.subheader("⚡ Features")
90
91     st.markdown("""
92     - 📚 Study Assistance
93     - 🤖 AI-Powered Responses
94     - ⚡ Fast Answers
95     - 🎯 Concept Explanations
96     - 📖 Learning Support
97     """)
98
99     st.divider()
100
```

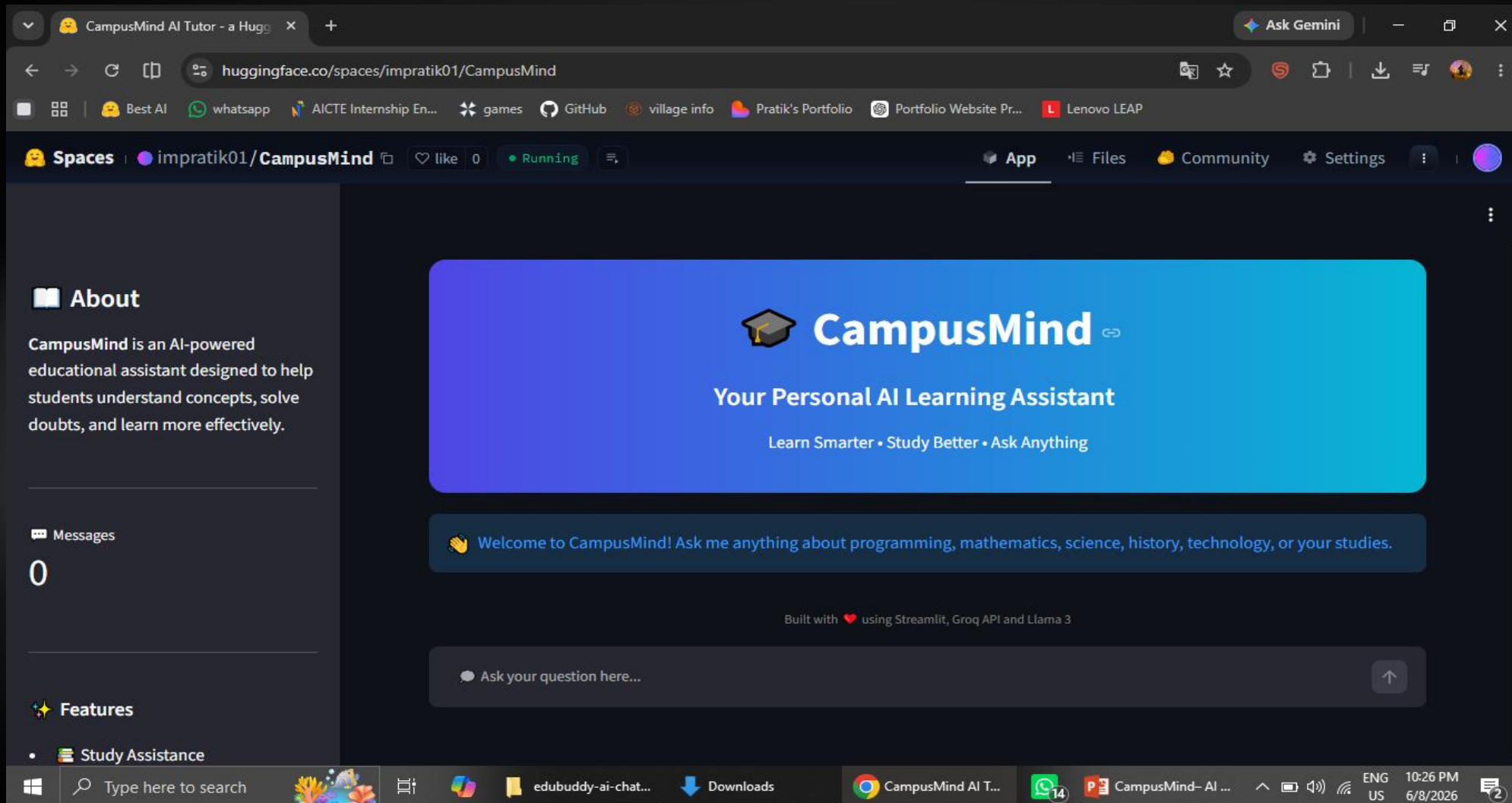


4. Results (Code Snippet 3)

```
E: > edubuddy-ai-chatbot-main > app.py > ...
112 | | st.markdown(message["content"])
113
114 # Chat input
115 if prompt := st.chat_input("👉 Ask your question here..."):
116
117     # User message
118     st.session_state.messages.append(
119         {"role": "user", "content": prompt}
120     )
121
122     with st.chat_message("user"):
123         st.markdown(prompt)
124
125     # Assistant response
126     with st.chat_message("assistant"):
127         with st.spinner("🧠 Thinking..."):
128
129             try:
130                 messages = []
131
132                 for m in st.session_state.messages:
133                     messages.append(
134                         {
135                             "role": m["role"],
136                             "content": m["content"]
137                         }
138                     )
139
140                 chat_completion = client.chat.completions.create(
141                     messages=messages,
142                     model="llama-3.1-8b-instant",
143                 )
144
```



4. Results (AI Chatbot - EduBuddy Main Interface)



4. Results (AI Chatbot - EduBuddy Response 1)

Spaces

impratik01/CampusMind

like 0

Running

Logs

AppFilesCommunitySettings

About

CampusMind is an AI-powered educational assistant designed to help students understand concepts, solve doubts, and learn more effectively.

Messages

0

Features

- Study Assistance
- AI-Powered Responses
- Fast Answers
- Concept Explanations
- Learning Support

CampusMind

Your Personal AI Learning Assistant

Learn Smarter • Study Better • Ask Anything

Welcome to CampusMind! Ask me anything about programming, mathematics, science, history, technology, or your studies.

Explain chemistry in details in 500 words

Introduction to Chemistry

Chemistry is the scientific study of the composition, properties, and reactions of matter. It is a fascinating field that deals with the interaction of atoms and molecules, and it has numerous applications in various fields such as medicine, technology, and environmental science. In this explanation, we will delve into the basics of chemistry, its branches, key concepts, and some of its most significant applications.

Branches of Chemistry

Chemistry is a vast and diverse field that can be divided into several branches, including:

Ask your question here...

4. Results (AI Chatbot - EduBuddy Response 2)

Spaces | impratik01/CampusMind

like 0

Running

Logs

App | Files | Community | Settings

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how to solve $2x + 5 = 11$?

To solve the equation $2x + 5 = 11$, we need to isolate the variable 'x'.

Here's the step-by-step process:

1. Subtract 5 from both sides of the equation: $2x + 5 - 5 = 11 - 5$ $2x = 6$

2. Now, divide both sides by 2 to isolate 'x': $2x / 2 = 6 / 2$ $x = 3$

So, the value of x is 3.

Ask your question here...

↑

4. Results (AI Chatbot - EduBuddy Response 3)

iMind

like 0

Running

Logs

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 explain newton's 3rd law

 **Newton's 3rd Law: The Law of Action and Reaction**

Newton's 3rd Law of Motion, also known as the Law of Action and Reaction, is a fundamental principle that describes the relationship between a body and the forces that act upon it. It states:

"Every action has an equal and opposite reaction."

In other words, when an object exerts a force on another object, the second object will respond with a force of equal magnitude but opposite direction. This law applies to all interactions between objects, whether it's a collision, a push, or even a gravitational pull.

Key Points:

Ask your question here...

↑

5. Project Links

🤖 Hugging Face Deployed (Live Demo):

<https://huggingface.co/spaces/impratik01/CampusMind>

🤖 Github: <https://github.com/impratik011/CampusMind-ai-chatbot>

6. Future Scope

Possible improvements for the future:

1. **Voice input** - Allow students to ask questions by speaking
2. **Multiple languages** - Support Hindi, Marathi, and other Indian languages
3. **Mobile app** - Create Android/iOS version for easier access
4. **Quiz generation** - AI can test students after teaching a topic
5. **Progress tracking** - Save chat history and track learning progress

7. Conclusion

CampusMind successfully demonstrates how AI can help achieve SDG 4 (Quality Education) by providing free, 24/7 personalized learning support to students who lack access to tutors. The chatbot uses Groq's Llama 3 model to answer questions in simple, easy-to-understand language. This project shows that AI can be a powerful tool for making education more accessible and equitable.

THANKS
FOR WATCHING

